

BUSINESS PLANNING AND TARIFF SETTING TECHNICAL ASSISTANCE

Appendix A. Tagbilaran 5-Year Business Plan

This is a **PRELIMINARY Business Plan with INDICATIVE tariffs** developed by the waterworks participants during their one-day workshop. The local waterworks is encouraged to review, update and complete it, and use it in their day-to-day operations.



NOVEMBER 29, 2025

Table of Contents

List of Tables	i
List of Figures	ii
Acronyms List	iii
1	1
1.1	1
1.2	1
1.3	2
1.4	2
1.5	4
1.6	5
2	9
2.1	9
2.2	10
2.3	15
2.4.1	16
2.4.2	20
2.4.3	21
2.5	22
2.5.1	23
3	25
References	27
Pictures	28

List of Tables

Table 1. Details of Water Service Coverage per Barangay	3
Table 2. Overview of Operations of Water Service Providers in Tagbilaran City	3
Table 3. Tagbilaran City Waterworks - Technical Data	7

Table 4. Tagbilaran City Waterworks - Commercial Data	7
Table 5. Tagbilaran City Waterworks - Financial Data	7
Table 6. Tagbilaran City Waterworks - Organizational Data	8
Table 7. Tagbilaran City Waterworks - Revised Vision, Mission, Values	9
Table 8. Sample Asset Registry Considering Three Pump Stations from TCWS Asset Inventory	10
Table 9. Description of Criticality Rating	11
Table 10. Sample Criticality Calculation for an Asset Management Plan	12
Table 11. Actual Criticality Rating for Three Pump Stations in Heat Map Colors	13
Table 12. Tagbilaran City Waterworks - Asset Management Plan	14
Table 13. Tagbilaran City Waterworks - SWOT Analysis	15
Table 14. Summary of 5-year CAPEX	20
Table 15. Summary of 5-year OPEX	20
Table 16. Tagbilaran City Waterworks - Calculated Should-be Water Tariffs	21
Table 17. Tagbilaran City Waterworks - Calculated Water Tariff for 2030 to 2034	21
Table 18. . Tagbilaran City Waterworks - Water Bill Calculation Across the WSPs at 10, 20 and 30 cubic meter consumption	21
Table 19. Key Performance Indicator Monitoring for Tagbilaran City Waterworks	22
Table 20. Population, Household and Tourist Projection for Tagbilaran City	24
Table 21. Water Service Connections Projection for Tagbilaran City	24
Table 22. Total Demand for Tagbilaran City	25
Table 23. NRW Projection for Tagbilaran City	25
Table 24. Total Unconstrained Demand for Tagbilaran City	25
Table 25. Net Demand for Tagbilaran City which will be served by TCWS	26

List of Figures

Figure 1. City Map of Tagbilaran	1
Figure 2. Coverage of All WSPs in Tagbilaran City	2
Figure 3. Tagbilaran City Waterworks - Water Coverage Map	5
Figure 4. Tagbilaran City Waterworks - Water Network Map	6
Figure 5. Tagbilaran City Waterworks - Water Availability Map	6
Figure 6. Tagbilaran City Waterworks Table of Organization	8
Figure 7. A Sample Heat Map and Criticality Scoring	13
Figure 8. Action Plan "Waterfall"	16
Figure 9. Pipe Replacement/Reinforcement -Tagbilaran City	18
Figure 10. Expansion of Coverage - Tagbilaran City	19
Figure 11. Main lines Project - Tagbilaran City	19

Acronyms List

ADD	Average Daily Demand
BWUI	Bohol Water Utilities Inc.
CAPEX	Capital Expenditures
CBMS	Community Based Monitoring System
CoF	Consequence of Failure
DMA	District Metering Areas
HP	horsepower
JICA	Japan International Cooperation Agency
JMC	Joint Memorandum Circular
KPI	Key Performance Indicator
LCE	Local Chief Executive
LEE	Local Economic Enterprise
LPCD	liters per capita per day
MLD	million liters per day
NEDA	National Economic Development Authority
NRW	Non-Revenue Water
OPEX	Operating Expenditures
PNSDW	Philippine National Standards for Drinking Water
PoF	Probability of Failure
PPDO	Provincial Planning and Development Office
SWOT	Strengths, Weaknesses, Opportunities, Threats
TA	Technical Assistance
TCWS	Tagbilaran City Waterworks
USAID	United States Agency for International Development
USAID-SURGE	United States Agency for International Development - Strengthening Urban Resilience for Growth with Equity
VMV	Vision, Mission, Values
WB	World Bank
WSC	water service connection
WSP	water service provider
WTP	water treatment plant

1.3 Water Supply

The city waterworks currently depends on deep wells, while private WSPs use both deep wells and surface water. Average production figures, in million liters per day (MLD), are as follows: Tagbilaran City Waterworks (TCWS) – 7.6 MLD (based on Jan–Jun 2025 data); Bohol Water Utilities, Inc. (BWUI) – 17.3 MLD (based on Jan–July 2025 data); and Richli Water – a total of 15 MLD from its Cortes and Loboc water treatment plants.

1.4 Water Coverage

Tagbilaran City, which has 15 barangays, is served by the three WSPs: TCWS, BWUI and Richli Water. All WSPs have Level III water systems¹. The map below illustrates where these WSPs are currently serving within the city.

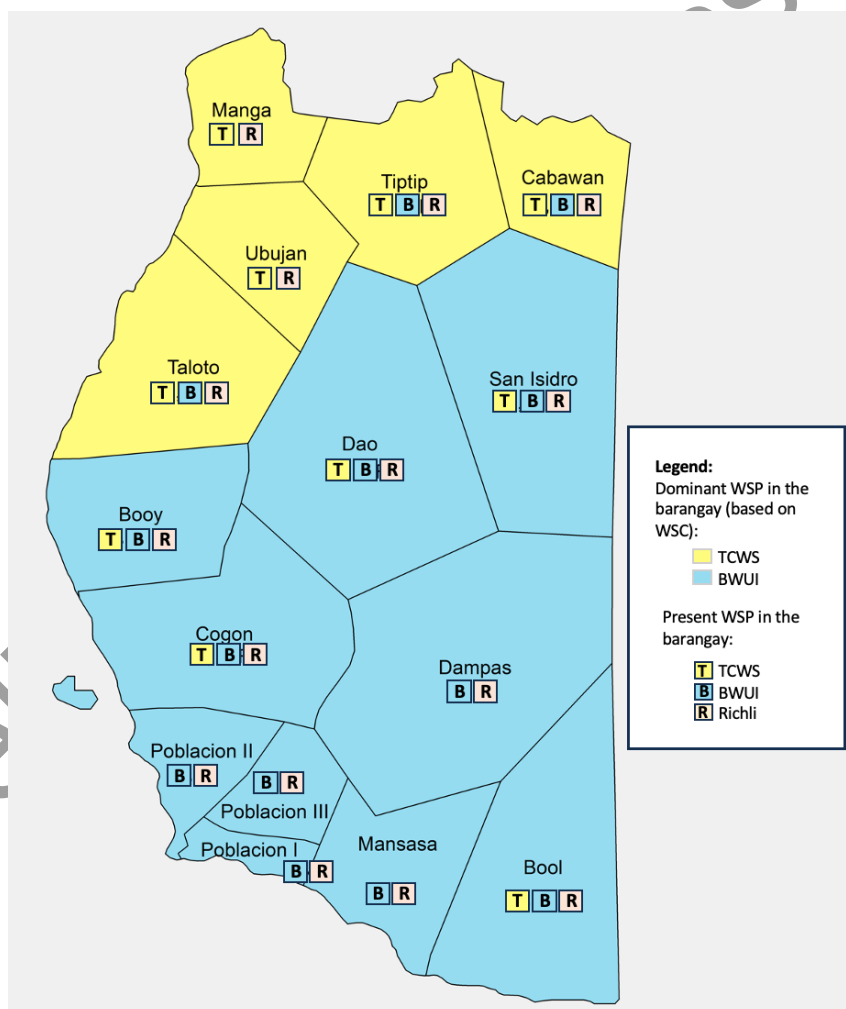


Figure 2. Coverage of All WSPs in Tagbilaran City

¹ The system includes a source, a reservoir, a piped distribution network and individual household taps (from the NEDA Board Resolution No. 12, Series of 1995.)

Table 1 below provides the details of water service coverage per barangay vis-à-vis their service provider.

Table 1. Details of Water Service Coverage per Barangay

	Barangay	Total Population (based on projected 2024 pop'n)	Projected HH (HH size: 4.5)	Prevailing WSP in the barangay	Barangay Presence			Total WSC			% of population served		
					Waterworks	BWUI	Richli	TCWS	BWUI	Richli	TCWS	BWUI	Richli
1	Bool	7,441	1,654	BWUI	served	served	served	114	1,401	no data	7%	77%	no data
2	Booy	10,698	2,377	BWUI	served	served	served	264	930	no data	11%	37%	no data
3	Cabawan	2,326	517	TCWS	served	served	served	231	7	no data	44%	0%	no data
4	Cogon	16,983	3,774	BWUI	served	served	served	21	3,385	no data	1%	75%	no data
5	Dampas	9,709	2,158	BWUI	unserved	served	served	-	1,994	no data	0%	82%	no data
6	Dao	11,446	2,544	BWUI	served	served	served	903	1,626	no data	35%	60%	no data
7	Manga	8,082	1,796	TCWS	served	unserved	served	1,333	-	no data	74%	0%	no data
8	Mansasa	6,981	1,551	BWUI	unserved	served	served	-	1,201	no data	0%	73%	no data
9	Poblacion I	3,131	696	BWUI	unserved	served	served	-	492	no data	0%	64%	no data
10	Poblacion II	4,644	1,032	BWUI	unserved	served	served	-	1,253	no data	0%	87%	no data
11	Poblacion III	5,391	1,198	BWUI	unserved	served	served	-	1,134	no data	0%	80%	no data
12	San Isidro	6,811	1,514	BWUI	served	served	served	200	1,100	no data	13%	70%	no data
13	Taloto	6,831	1,518	TCWS	served	served	served	831	446	no data	54%	26%	no data
14	Tiptip	5,922	1,316	TCWS	served	served	served	984	71	no data	74%	5%	no data
15	Ubujan	5,952	1,323	TCWS	served	unserved	served	1,203	-	no data	90%	0%	no data
TOTAL		112,348	24,966					6,084	15,040	-	24%	53%	

Table 2 below provides an overview of the operations and performance of the three WSPs.

Table 2. Overview of Operations of Water Service Providers in Tagbilaran City

	TCWS	BWUI	Richli
Served barangays in Tagbilaran City	10 out of 15 (Bool, Booy, Cabawan, Cogon, Dao, Manga, San Isidro, Taloto, Tiptip, Ubujan)	13 out of 15 (Bool, Booy, Cabawan, Cogon, Dampas, Dao, Mansasa, Poblacion I,II,III, San Isidro, Taloto, Tiptip)	all 15 (Bool, Booy, Cabawan, Cogon, Dampas, Dao, Manga, Mansasa, Poblacion I,II,III, San Isidro, Taloto, Tiptip, Ubujan)
Billed volume (MLD)	Jan to June 2025: 5.26 MLD	Jan to July 2025: 15.56 MLD	No data provided
Production volume (MLD)	Jan to June 2025: 7.66 MLD	Jan to July 2025: 17.33 MLD	No data provided
NRW (%)	31%	7%	10%
Water availability (hours/day)	22 to 24 hours	23 hours	24 hours
Water supply sources	Deep wells	Deep wells	Surface water (5 MLD WTP in Loctob, Loboc; and 10 MLD WTP in Cortes)
Water supply treatment	chlorination	chlorination	filtration, chlorination

	TCWS	BWUI	Richli
Water service connections (total count)	6,084 (as of June 2025)	15,040 (as of July 2025)	no data provided, but majority of its ~15,000 wsc
Average monthly consumption (cu.m./conn./month)	26.1	31.3	no data provided
Collection Efficiency	No data provided	No data provided	No data provided, but high collection efficiency
Water quality	No customer complaints	for deep wells: salt intrusion in wells near coastal areas, seasonal turbidity, some wells have iron and bacterial presence	No data provided
Demand gap	Not provided	current: ~10 MLD (for existing coverage area)	No data provided
Minimum water tariff (PhP/cu.m.)	residential and institutional: 0-10 cu.m.: PhP62.00 commercial: PhP23/cu.m.	residential and institutional: 0-5 cu.m.: PhP100.00 6-10 cu.m.: PhP20/cu.m. commercial: 0-15 cu.m.: PhP1,107.00	residential and institutional: 0-5 cu.m.: PhP218.96 6-10 cu.m.: PhP43.79 commercial: 0-8 cu.m.: PhP792.00

1.5 Sanitation²

Currently, Tagbilaran City is served by various wastewater management providers to maintain environmental standards and public health. These providers are responsible for ensuring that wastewater from residential, commercial, and industrial sources is appropriately treated and disposed of in compliance with national and local regulations. Industrial and commercial entities, such as large resort hotels discharging 100 cubic meters (cu.m.) per day, are required to install their treatment facilities for their wastewater.

In terms of septage management, there are planned facilities to address the sanitation needs in Tagbilaran City. A septage treatment plant (SpTP) in Tagbilaran City, located near the bus terminal, is currently being developed. It has a capacity of 24 cu.m./day. It is a pilot SpTP under a Japan International Cooperation Agency (JICA) project. Another proposed SpTP has a capacity of 25-100 cu.m./day to be located at Brgy. Tiptip. It is intended to serve Tagbilaran City only.

²AWSPSA - Appendix A.1: The Loboc Cluster LGUs and the Status of their WSS and WSPs. (2025). DPWH.

Data from the Bohol Provincial Planning and Development Office (PPDO) as of May 2024 indicate 103 percent of households in Tagbilaran has basic sanitation facilities; 103 percent has pour/flush toilet connected to a septic tank; and 103 percent use safely managed sanitation services.

1.6 Tagbilaran City Waterworks (TCWS)

As of June 2025, TCWS serves 10 of the city's 15 barangays through 6,084 water service connections, with Barangays Manga and Ubujan exclusively reliant on TCWS (Figure 3). Its network map is provided in Figure 4. Average water availability across the service area is 22 to 24 hours per day (Figure 5). The system sources water entirely from groundwater and operates with 11 regular personnel, reporting directly to the mayor. Despite this, Tagbilaran faces an 11.7 MLD supply gap between current waterworks output and projected unconstrained demand.

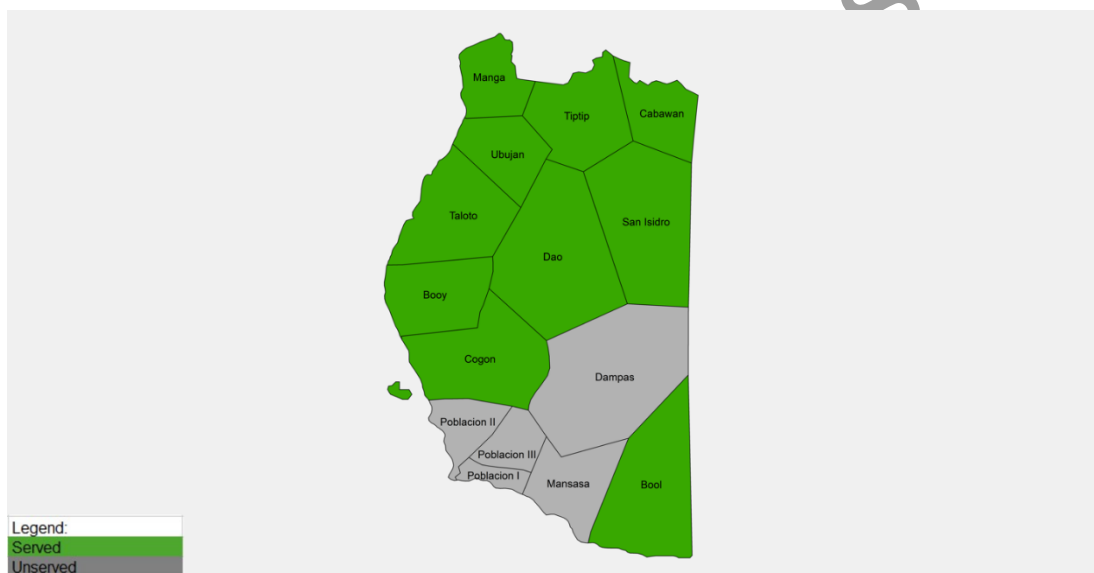


Figure 3. Tagbilaran City Waterworks - Water Coverage Map

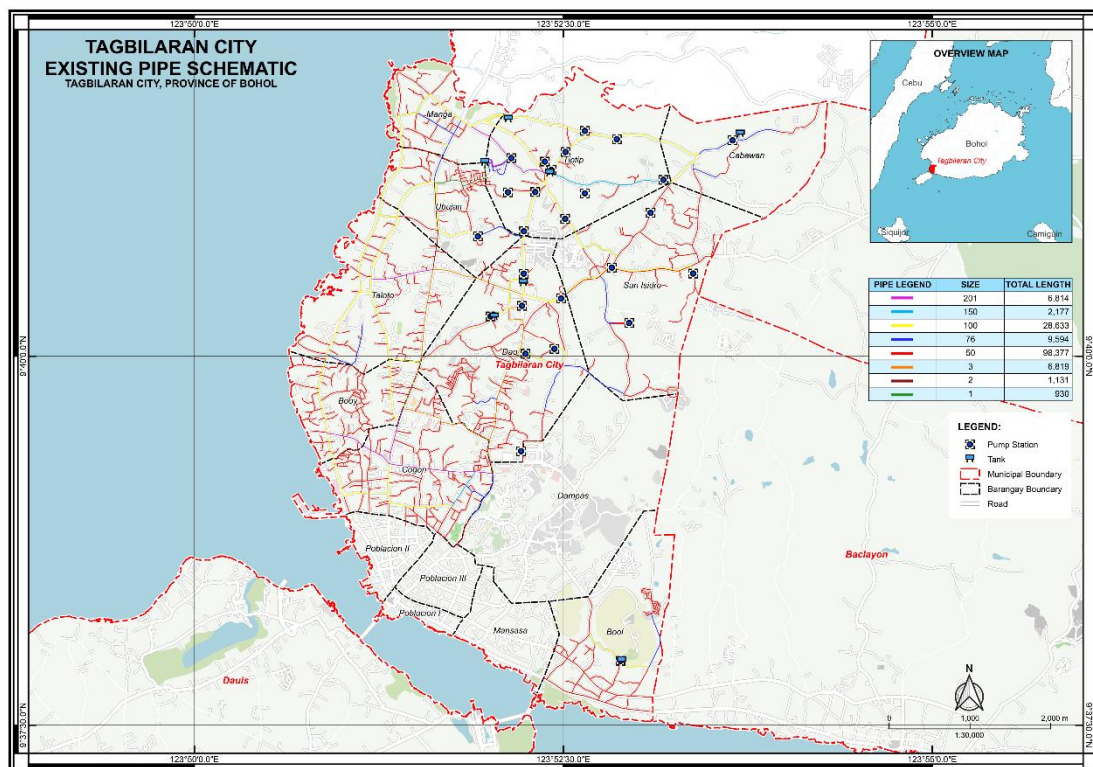


Figure 4. Tagbilaran City Waterworks - Water Network Map

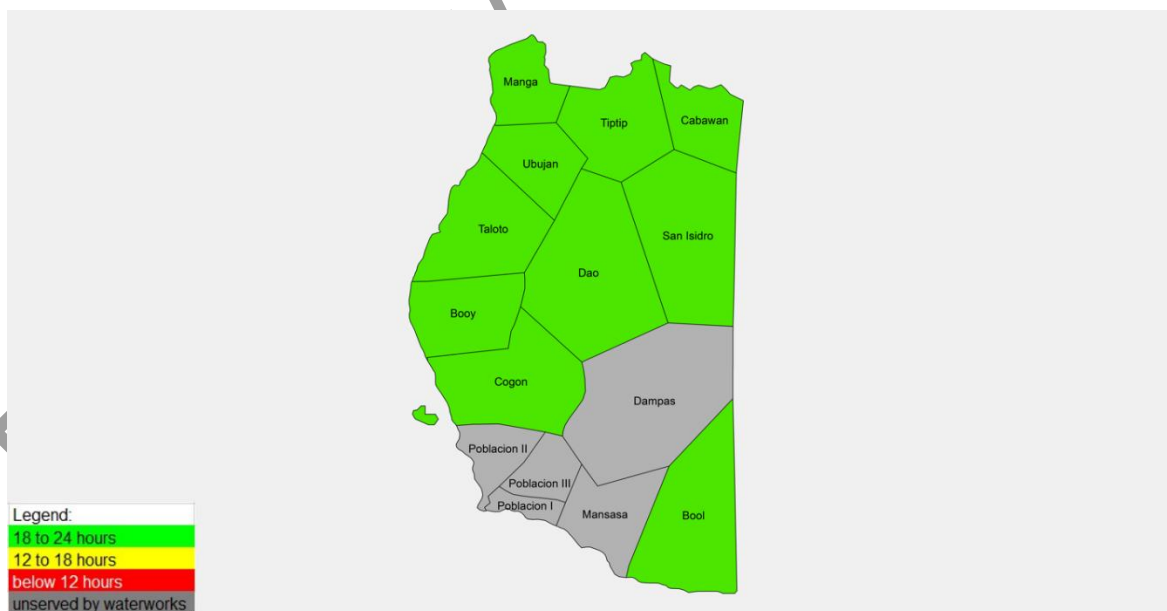


Figure 5. Tagbilaran City Waterworks - Water Availability Map

Tables 3 to 6 provide the Technical Data, Commercial Data, Financial Data and Organizational Data of Tagbilaran City Waterworks.

Table 3. Tagbilaran City Waterworks - Technical Data

Served barangays	10 out of 15 barangays
Total population served	approx. 27,000 people
Billed volume (in MLD)	5.26 MLD (Jan to June 2025 average)
Production volume (in MLD)	7.66 MLD (Jan to June 2025 average)
Non-revenue Water (in %)	31% (calculated)
Water availability (hours/day)	22 to 24 hours
Water quality	No customer complaints
Average pressure	No data provided
Current assets	Pumping stations, reservoirs, and pipelines
Water supply source	Groundwater
Water supply treatment	Chlorination

Table 4. Tagbilaran City Waterworks - Commercial Data

Water Service Connection	as of August 2025
Residential	6,017
Commercial	34
Institutional	33
Industrial	0
Bulk	0
TOTAL	6,084
Average monthly consumption (cu.m./month)	26.1

Table 5. Tagbilaran City Waterworks - Financial Data

Annual Revenue (PhP/year)	Not provided
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Annual Water Sales (PhP/year)	2024: 22,887,685.65 Jan-Jun 2025: 12,036,951.99
Collection Efficiency (% Ytd)	No calculated data
Annual Operating Expenses (PhP/year)	Not provided
Minimum Tariff (PhP/cu.m.)	residential and institutional: 0-10 cum: 62.00/cu.m. commercial: 23.00/cu.m.
Average Tariff (PhP/cu.m.)	13.76

Table 6. Tagbilaran City Waterworks - Organizational Data

Staff Count	Male	Female	Total
Regular	8	1	9
Job Order	10	5	15
Casual			
Total	18	6	24

The details on how the waterworks is organized is shown in Figure 6.

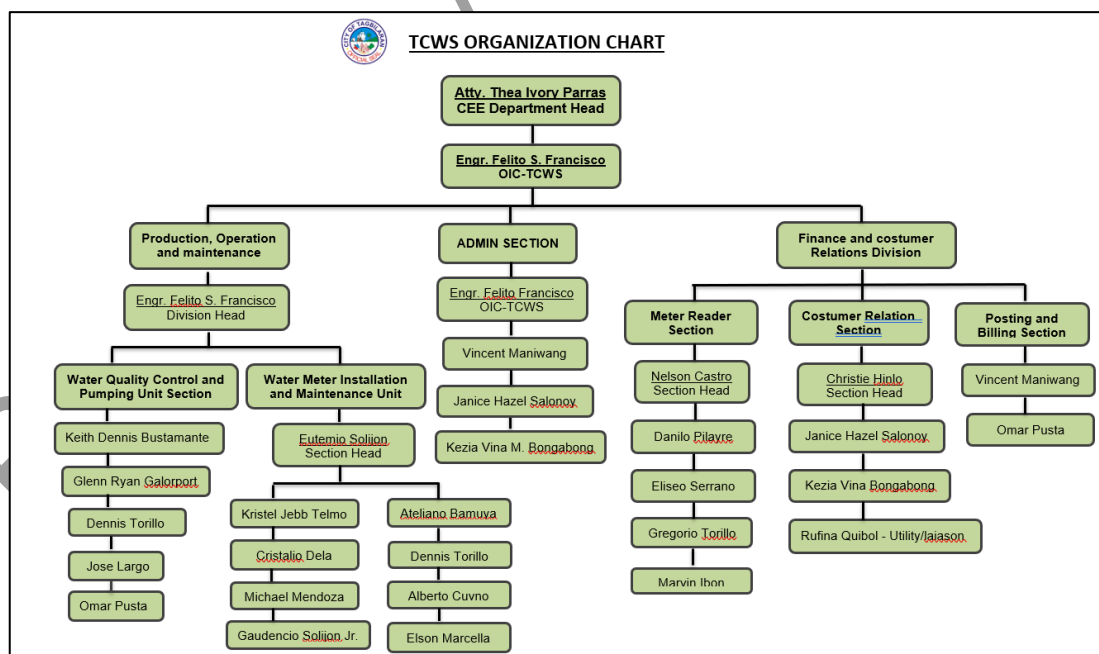


Figure 6. Tagbilaran City Waterworks Table of Organization

Source: Tagbilaran City Waterwork

2 5-Year Business and Tariff Plan

2.1 Vision, Mission and Core Values (VMV)

Tagbilaran City Waterworks has an existing vision, mission and values which reads as follows:

Vision	To manage our organization and water resources to meet evolving regulatory requirements, water supply needs, and customer expectations in the future.
Mission	Our mission is to provide our customers with safe and reliable water in compliance with regulatory requirements. We strive to treat every customer fairly and respectfully, and to address their needs quickly and effectively.
Values	W-isdom A-accomplishment T-eamwork E-nergy R-esponsibility W-elcoming O-bedience R-espect K-indness S-ervice

But the waterworks came up with a revised VMV (Table 7) during their business planning exercise in November 2025 after evaluating it using some parameters by David & David (2017).

TCWS re-wrote its vision by responding to the following questions: Where are we going moving forward? What do we want to achieve in the future? What kind of future state do we envision? What impact do we want to leave the community 10-20 years from now?

It also improved its existing mission statement by answering the following: What do we do today? Who do we serve? What are we trying to accomplish?

Finally, it reevaluated its 10 values and retained those that will help it achieve its vision. The waterworks had to respond to the following questions to articulate its values: What do we stand for? What behaviors are important to us? How will we conduct our activities to achieve our vision/mission? How do we treat the members of our own organization and community?

Table 7. Tagbilaran City Waterworks – Revised Vision, Mission, Values

Vision	We want to become a world-class, sustainable waterworks by 2035. We want to achieve highly potable, 24/7 water supply. We see a more effective waterworks company capable of providing reliable and safe/clean water to every household in Tagbilaran City.
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Mission	Our mission is to provide our customers in 10/15 barangays in Tagbilaran City with safe and reliable water. Our employees are capable and highly trained. We strive to treat every customer fairly and respectfully and address their needs quickly and effectively.
Values	The following values will help us achieve our vision and guide our mission: Teamwork Community-based Service Respect Dedication Adaptiveness Resilience Innovation

**The VMV will be reviewed and re-evaluated by Tagbilaran Waterworks.*

2.2 Asset Management Plan

TCWS needed to have a thorough understanding of the water system's infrastructure and its capabilities or an asset management plan. A simple asset management plan answers five core questions: What do we own? What service do we need to deliver? What are our most important assets? What do we need to do to keep them operating? How do we pay for it?

To guide the exercise, the study team introduced four primary reasons why assets fail: mortality, capacity, level of service, and financial efficiency (U.S. EPA, 2016).

- **Mortality** occurs when an asset physically fails or can no longer perform its intended function.
- **Capacity** means the asset remains operational but cannot meet the required output.
- **Level of service** refers to an asset that works but fails to deliver the expected service quality.
- **Financial efficiency** arises when operating costs become uneconomical.

The study team also presented the concept of an asset registry, the foundation of any asset management plan. During the exercise, the waterworks team learned how to create a basic registry, apply proper tagging, and record essential data for each asset. Table 8 below shows a sample registry, using the actual assets of TCWS including pertinent data: location, capacity, condition and year constructed.

Table 8. Sample Asset Registry Considering Three Pump Stations from TCWS Asset Inventory

Asset Type	Location	Specifications/ Capacity	Condition (poor, fair, good)	Year Constructed
Pumping Stations	San Roque Road, Tiptip (near Cuyunos Residence)	5HP	good	2021

Asset Type	Location	Specifications/ Capacity	Condition (poor, fair, good)	Year Constructed
	7 Darunday St. Dao (near Cornell Residence)	7.5HP	bad, encounters, breakdowns, frequent maintenance	2021
	12 Manga-Tiptip-Cabawan Road, Tiptip (near Serrano Residence)	15HP	fair	2022

HP = horsepower

The concept of criticality was also introduced to the Tagbilaran Waterworks personnel. Criticality is defined as the measure of risk associated with an asset, calculated as the Probability of Failure (PoF) × Consequence of Failure (CoF) (Betgeri, Vatyala & Matthews, 2024). For effective planning, the waterworks must identify the assets that are most likely to fail; assess the impact of such failures; and determine the optimal timing for replacement or procurement of critical spares.

Understanding the principles of asset inventory, criticality, and asset management is essential for maximizing financial resources while minimizing downtime. By applying criticality analysis, projects and asset renewals can be prioritized to ensure the efficient use of limited funds and personnel (Moss & Woodhouse, 1999).

The matrix in Table 9 below serves as the basis for determining asset criticality ratings. Each asset was scored on two dimensions: PoF and CoF, using a scale of 1 to 5, where 1 = lowest and 5 = highest. While other scoring systems can be applied, the 1–5 scale is widely used in industry. The maximum criticality score is 25 (PoF × CoF), with higher scores indicating greater risk.

Table 9. Description of Criticality Rating

Risk Score	Metrics for Likelihood of Failure	Metrics for Consequence of Failure
1	Highly unlikely to fail within the year	0-100 Service connections affected
2	Unlikely to fail within the year	101-300 Service connections affected
3	Probable to fail within the year	301-600 Service connections affected
4	High probability of failure	601-1000 Service connections affected
5	Certain to fail within the year	More than 1000 service connections affected

To illustrate, a criticality rating assessment was conducted for three pumping stations of TCWS (Table 10).

Table 10. Sample Criticality Calculation for an Asset Management Plan

Pump Stations	Probability of Failure (PoF)	Consequence of Failure (CoF)	Criticality (PoF x CoF)
San Roque Road, Tiptip (near Cuynos Residence)	3	2	6
7 Darunday St, Dao (Near Cornell Residence)	3	3	9
12 Manga-Tiptip-Cabawan Road, Tiptip (Near Serrano Residence)	2	5	10

The urgency of asset renewal can be visualized by plotting criticality scores on a heat map, as shown in Figure 7. Assets are positioned based on their PoF and CoF ratings. For example, an asset with a PoF of 5 (certain failure) and a CoF of 5 (catastrophic impact) will appear in the upper-right red zone. This means that immediate replacement of the asset is required.

- **Red Zone:** High-risk assets requiring urgent renewal or replacement.
- **Yellow Zone:** Moderate-risk assets that need budget allocation for replacement or critical spares in the next planning cycle.
- **Green Zone:** Low-risk assets, generally excluded from the upcoming replacement budget unless their status changes in future updates.

This visual tool helps prioritize investments and maintenance actions based on risk severity.³

³ For further references on how to assess asset risks through heat maps, you may refer to the publication “Risk Heat Map: An Updated Risk Assessment Tool” ([Risk Heat Map An Updated Risk Assessment Tool.pdf](#)).

IMPACT	Catastrophic (5)	5	10	15	20	25
	Significant (4)	4	8	12	16	20
	Moderate (3)	3	6	9	12	15
	Low (2)	2	4	6	8	10
	Negligible (1)	1	2	3	4	5
		Improbable (1)	Remote (2)	Occasional (3)	Probable (4)	Frequent (5)
		LIKELIHOOD				

Figure 7. A Sample Heat Map and Criticality Scoring

Source: <https://www.techtarget.com/searchbusinessanalytics/definition/heat-map>

Plotting the calculated criticality rating in Table 10 in the heat map would yield the following results, as shown in Table 11.

Table 11. Actual Criticality Rating for Three Pump Stations in Heat Map Colors

No.	Pump Station	Probability of Failure (PoF)	Consequence of Failure (CoF)	Criticality Score (PoF x CoF)
1	San Roque Road, Tiptip (Near Cuynos Residence)	3	2	6
2	7 Darunday St, Dao (Near Cornell Residence)	3	3	9
3	12 Manga-Tiptip-Cabawan Road, Tiptip (Near Serrano Residence)	2	5	10

Probability of Failure (PoF)	Certain					
	Highly Probable					
	Probable		1	2		
	Unlikely					3
	Highly Unlikely					
Criticality = PoF X CoF		0-100 Service connections affected	101-300 Service connections affected	301-600 Service connections affected	601-1000 Service connections affected	More than 1000 Service connections affected
		Consequence of Failure (CoF)				

Table 12 below presents a sample asset management plan prepared by TCWS and the study team. The last two entries were initiated during the workshop to demonstrate the process. TCWS is encouraged to continue this exercise for all remaining assets to build a comprehensive asset management plan. TCWS should be updating this asset management plan yearly.

Table 12. Tagbilaran City Waterworks - Asset Management Plan

Asset Type	Asset ID	Location	Condition	Remaining Useful Life	Criticality /Risk Rating	Recommended Action	Estimated Replacement Cost (in PhP)
Pump	PS-001-2022	San Roque Road, Tiptip (near Cuynos Residence)	good	2	6	plan for replacement in two years	300,000
Pump	PS-006-2020	7 Darunday St, Dao (near Cornell Residence)	breakdowns, frequent maintenance	0	9	for immediate replacement	600,000

Pump	PS-002-2022	7 Darunday St, Dao (near Cornell Residence)	fail	0	10	immediate replacement	600,000
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2.3 SWOT (Strengths, Weaknesses, Opportunities, Threats) Analysis

The study team then facilitated a SWOT Analysis exercise. TCWS was given examples and then asked to look at its strengths and weaknesses as a water service provider. This includes evaluating where it was good or weak at; how reliable or unreliable their water system is; how well maintained or not are their assets; how their customers see them and areas for improvement. The WSP was then asked to evaluate the external environment and identify opportunities and threats. For opportunities, the following questions were raised to trigger the discussion: *Which unserved areas can we expand to? Where are the high growth development areas in the city? How can we partner with the province? How can we maximize existing septage facilities? What technology solutions can we adopt to address some of our existing issues?* For threats, the waterworks was asked the following: *How stable or volatile is our electricity cost? What is the risk of our customers being unable to pay the water tariff? What is the threat from so many other water providers in the city? Are our water sources threatened by land-based pollution and/or salt water intrusion?*

TCWS came up with the following SWOT table during the exercise (Table 13).

Table 13. Tagbilaran City Waterworks - SWOT Analysis

Strengths	Weaknesses
- Committed and customer-friendly staff - Staff can work under pressure - Already under a local economic enterprise (LEE) and not funded by general funds - Installed chlorinators in all pumping stations - Responsive team (especially in emergency services) - Customer expectations are effectively managed - Strong management support - Online payment partners	- Multi-tasking of personnel (organizational structure) - Old, galvanized iron pipes (parts of Brgys. Tiptip, Mangga, Ubujan) - Need training for technical, finance personnel - Low tariff - Lack of training for billing and collection, management training, water safety plans - Lack of water supply - Lack of preventive maintenance - Negative bottom line - Handling negative criticism - Lack of decision-making - Water quality monitoring and compliance (10% failure in sampling)

Opportunities	Threats
- Availability of bulk water supply in a few years' time - Large growth potential – many investors - Non-government organizations and other private organizations (i.e. USAID, Maynilad) who help improve water supply services - Other local water utilities' support (collaborative effort with BWUI)	- Climate change - Natural hazards (i.e. Typhoons, earthquakes) (but there are standby gensets in selected pumping stations) - Saltwater intrusion in water sources - Deep well sources of some private institutions / establishments / businesses

BWUI = Bohol Bulk Water Utilities Inc.

2.4.1 Action Plan

Following the first three steps and being aware of their weaknesses as a utility, the fourth step was to create an action plan to achieve their goals. To facilitate the exercise, the study team identified five critical goals for water utilities. These are improving billed volume, reducing NRW, improving collection efficiency, managing cost recovery, and improving after sales service. For each of the goals, the WSP was asked to identify one to two good strategies. And for each of the strategies, they had to identify specific projects (Figure 8).

Output 4a: Action Plan “Waterfall”

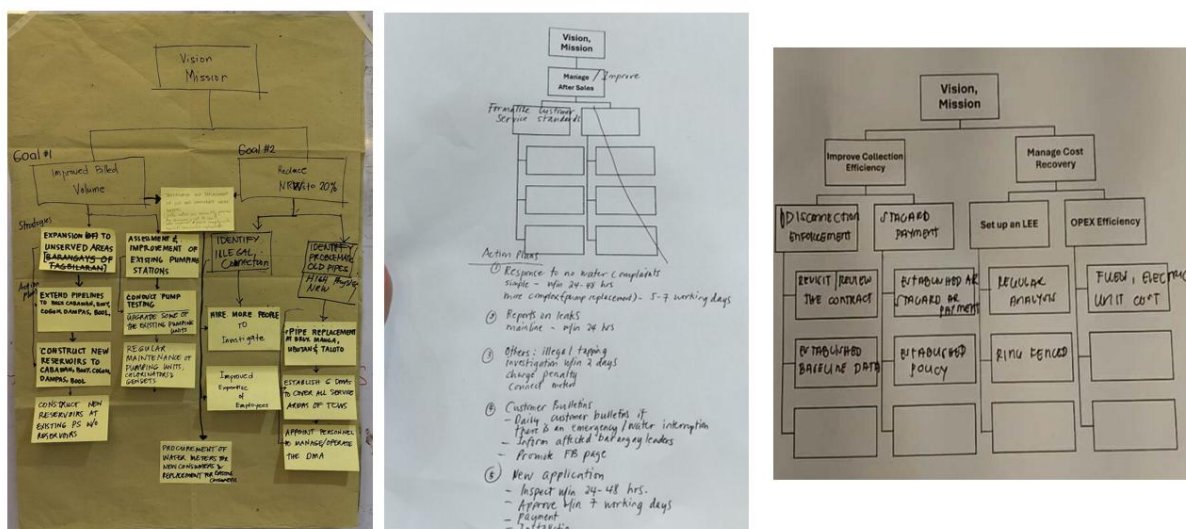


Figure 8. Action Plan “Waterfall”

TCWS produced the following preliminary action plans following the exercise.

Goal #1: Improve Billed Volume.

Strategy 1: Expansion to unserved areas

Action Plans	1. Extend pipelines to Brgys. Cabawan, Booy, Cogon, Dampas, Bool 2. Construct new reservoirs in the expansion areas in (1). 3. Construct new reservoirs at the existing pump stations without reservoirs.
Strategy 2: Assessment and improvement of existing PS	
Action Plans	1. Conduct pump testing. 2. Assess and upgrade some of the existing pump units. 3. Regularly maintain pump stations, chlorinators, and gensets.
Strategy 3: Identification and replacement of old and inaccurate meters	
Action Plans	1. Procurement of water meters for new consumers and meter recalibration or replacement for existing customers. <i>*Water meters are currently provided by consumers. Some fail to comply with the required or recommended replacements.</i>

Goal #2: Reduce NRW to 20%.

Strategy 1: Identify illegal connections	
Action Plans	1. Hire or designate more people to investigate illegal connections. 2. Improve the expertise of employees.
Strategy 2: Identify problematic old pipes/ high physical NRW	
Action Plans	1. Establish 6 DMAs to cover all service areas of TCWS. 2. Replace old pipes at Brgys. Manga, Ubujan, Taloto. 3. Appoint and train personnel to manage or operate established DMAs.

Goal #3: Improve Collection Efficiency.

Strategy 1: Disconnection enforcements	
Action Plans	1. Review or revisit the service contracts. 2. Establish baseline date.
Strategy 2: Staggered payments	
Action Plans	1. Establish staggered AR payment schedule. 2. Establish policies on accounts receivable.

Goal #4: Manage Cost Recovery.

Strategy 1: Set up an LEE.	
Action Plans	1. Ring-fence waterworks account. 2. Regular evaluation of job designations and descriptions.

Strategy 2: OPEX Efficiency	
Action Plans	1. Develop and implement energy efficiency measures, focusing on fuel and electricity utilization. 2. Regularly monitor fuel and electricity unit efficiency costs .

Goal #5: Manage/ Improve After Sales.

Strategy 1: Formalize Customer Service Standards	
Action Plans	1. Response to no water complaints simple – within 24-48 hours complex – 5 to 7 working days 2. Reports on leaks mainline – within 24 hours 3. Others: illegal tapping ▪ investigation within 2 days ▪ charge penalty ▪ connect meters 4. Customer Bulletins ▪ daily customer bulletins if there is an emergency/ water interruption ▪ inform affected barangay leaders ▪ promote Facebook page 5. New application ▪ Inspect within 24-48 hours ▪ Approve within 7 working days ▪ Payment then installation

Initial Proposed Projects

1. **Pipe Replacements and Reinforcements** in Brgys. Manga, Ubujan and Tiptip and Taloto with the following objectives: NRW Reduction, increase conveyance capacity and improve water availability and pressure. Implementation will be for Years 1 to 4 (Figure 8). Estimated total project cost is **PhP113.1 million**.

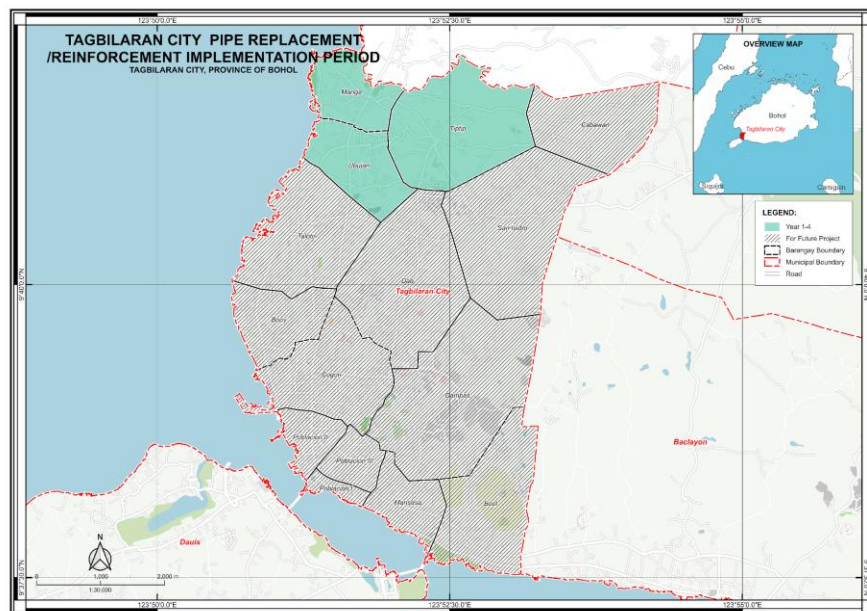


Figure 9. Pipe Replacement/Reinforcement - Tagbilaran City

2. **Expansion of coverage** in Brgy. San Isidro and Cabawan and Booy, Tampas and Bool to increase service connections and increase billed volume/ revenue. Implementation will be for Years 1 to 4 (Figure 9). Estimated total project cost is **Php22.9 million**.

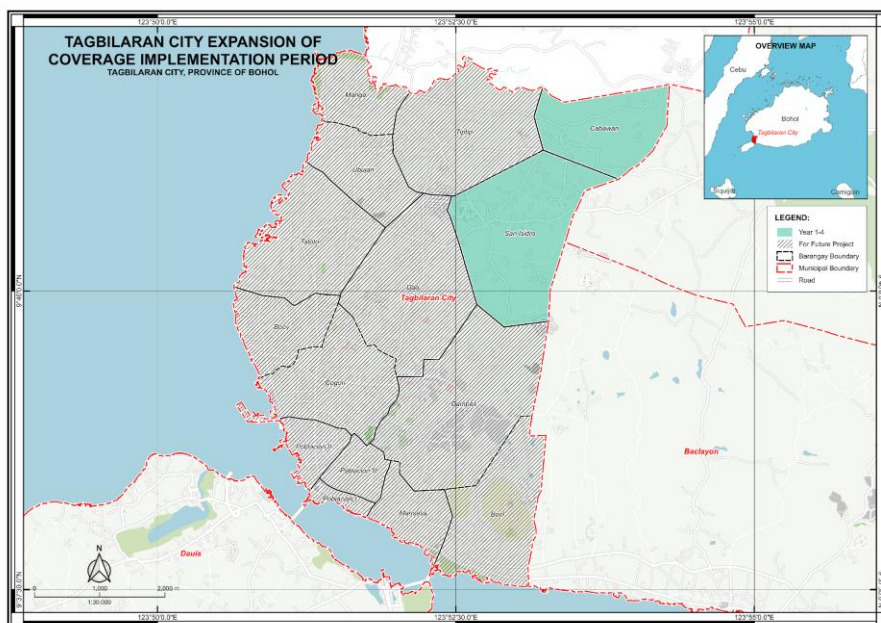


Figure 10. Expansion of Coverage - Tagbilaran City

3. **Main lines Project** for the acceptance of the increased supply coming from the bulk water project. These will be in Brgys. Dao (Year 2-4), Tiptip (Year 3-4), and San Isidro (Year 4) (Figure 10). Estimated total project cost is **Php119.4 million**.

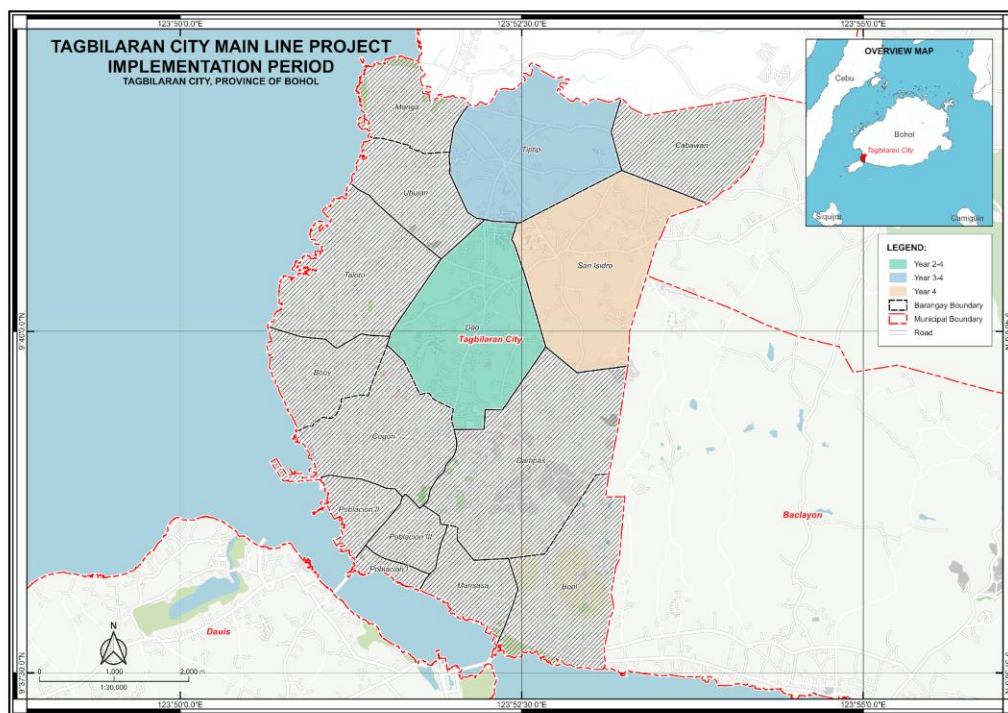


Figure 11. Main lines Project - Tagbilaran City

2.4.2 Financial Plan

Table 14 provides a summary of the major capital expenditures in the CAPEX plan.

Table 14. Summary of 5-year CAPEX

Project	Estimated CAPEX (in million PhP)				
	Construction Year				
	Y1 (2026)	Y2 (2027)	Y3 (2028)	Y4 (2029)	Y5 (2030)
Pipe Replacement/ Reinforcements	32.15	47.24	27.97	5.74	
Expansion of Coverage	7.18	7.79	5.97	1.99	
New Main Lines		26.11	44.4	48.91	
Total	39.33	81.14	78.34	56.64	

*It is assumed that construction period is from 2026 to 2029.

Meanwhile, the OPEX items considered in the financial plan are summarized in Table 15 below:

Table 15. Summary of 5-year OPEX

OPEX item	Estimated OPEX (in million PhP)				
	Operational Year				
	Y1 (2030)	Y2 (2031)	Y3 (2032)	Y4 (2033)	Y5 (2035)
Salaries	10.416	11.098	11.826	12.601	13.427
Power Cost	87.835	78.376	83.283	88.525	94.092
Chemical Cost	0	0	0	0	0
Bulk Water Purchase	132.790	112,847	114.216	115.610	117.028
Misc and Other O&M	11.552	10.116	10.466	10.836	11.227
Service Fee	24.259	21.243	21.980	22.757	23.577
TOTAL	266.854	233.682	241.784	250.331	259.353

*It is assumed that operations of the new system will commence in 2030.

2.4.3 Tariff Analysis

The proposed business plan for Tagbilaran is expected to benefit a total of 11,908 households by the end of 2039. It is found to be financially viable at the proposed rates of **PhP214.03** (Years 1 to 5) and **PhP267.52** (Years 6 to 10) for the first 10 cubic meters respectively. Currently, TCWS is charging PhP6.20 per cubic meter. In 2023, it should have already been charging PhP13.76 per cubic meter to recover OPEX only. During the workshop, using the most recent financial inputs, the accounting group calculated the applicable tariff to be at PhP17.68 for 2024.

Table 16. Tagbilaran City Waterworks – Calculated Should-be Water Tariffs

	2020 (Base Year)	2021	2022	2023	2024
OPEX (in PhP)	17,125,593	19,450,006	*21,015,606	*26,107,632	
Billed Volume (in cu.m.)	1,740,937	1,807,220	1,704,980	1,897,357	
Actual Tariff	6.20	6.20	6.20	6.20	
Should be tariff (in PhP/cu.m.)	9.83	10.76	12.32	13.76	17.68

*Calculated OPEX

The proposed **indicative** tariff for 2030 to 2034 remains affordable at 2 percent of the average income of the low-income group. Cash is positive throughout the period. With these findings, the proposed improvement plan is deemed financially viable.

Tables 17 and 18 show the calculated **indicative** water tariff and the calculation of water bills across different WSPs at different consumption level.

Table 17. Tagbilaran City Waterworks - Calculated Water Tariff for 2030 to 2034

Proposed Indicative Tariff w/ Bulk Supply (2030-2034)		Calculating the water bill at different consumptions	
Consumption	Calculated water bill (PhP/ month)	Consumption	Calculated water bill (PhP/ month)
Minimum Charge (0-10 cu.m.)	PhP214.03	At 10 cu.m.	214.03
11-20 cu.m.	PhP22.47/cu.m.	At 20 cu.m.	438.76
21-30 cu.m.	PhP23.60/cu.m.	At 30 cu.m.	674.72
31-40 cu.m.	PhP24.78/cu.m.		
41-50 cu.m.	PhP26.02/cu.m.		
51 up cu.m.	PhP27.32/ cu.m.		

Table 18. . Tagbilaran City Waterworks - Water Bill Calculation Across the WSPs at 10, 20 and 30 cubic meter consumption

Consumption	Calculated water bill at different consumptions (PhP/month)		
	TCWS (Proposed Indicative Tariff w/ Bulk Supply)	BWUI	Richli
10 cu.m.	214.03	200.00	437.91
20 cu.m.	438.76	436.00	898.21
30 cu.m.	674.72	724.00	1,379.81

2.5 KPIs

Finally, TCWS had to evaluate its business plan using **metrics and key performance indicators (KPIs)** from the draft JMC. The study team shared some practical steps on monitoring and tracking KPIs – from identifying the KPIs, setting targets, choosing a data collection method, creating a data management system, establishing the frequency of monitoring and reporting, to analyzing and using the data.

TCWS came up with the following preliminary KPI system.

Table 19. Key Performance Indicator Monitoring for Tagbilaran City Waterworks

	Standard (draft JMC)	Proposed Monitoring Method (source document, data)	Proposed Frequency of Monitoring and Reporting
Water coverage (%)	Based on what was specified in 5-year business plan	Source: monthly water service connections record	monthly

	Standard (draft JMC)	Proposed Monitoring Method (source document, data)	Proposed Frequency of Monitoring and Reporting
Water quality	Compliance with the PNSDW targets	Bacteriological test and physico-chemical test results	monthly (bacteriological test) and semi-annual (physico-chemical test)
Water availability (hours)	At least 18 hours	Source: customer complaints records	daily
Non-revenue water	20% or below	Source: production, actual billed volume data	monthly
Water pressure	At least 8 psi	Source: pressure monitoring readings	proposed: once every 2 weeks, off-peak and peak hours
Operating ratio	At least 1.00	Source: financial data	proposed: calculate monthly for OPEX related to waterworks
Revenue collection efficiency (%)	Not less than 95%	Source: financial data	monthly
Average tariff	-		
Staff/1000 connections ratio	1 regular staff: 100 connections	1:250 connections	annually
Connections metered (%)	-		
Average monthly consumption	-		
Number of desludged septic tanks/year	Based on set annual target		

*Staff to connection ratio of 1:250 is the current industry average.

2.5.1 Presentation to the Local Chief Executive

The preliminary business plan was presented to Mayor Jane Yap who joined the discussion online. The following were some of the mayor's comments:

1. There is a need to balance the number of weaknesses in relation to TCWS' strengths.

2. The low tariff in TCWS is its number one weakness and has always been a concern. But tariff increases need to be evaluated and balanced with better service.
3. There will be movements in TCWS because we need permanent heads for critical assets like pumping stations. The appointment of Engr. Felito to head TCWS has been good for the department and we have not had any major complaints that were not addressed. We will also be digitalizing the process and improving our billing system.
4. TCWS will not be sold to the private sector.

Preliminary Business Plan

3 Water Demand Analysis

This section on water demand was not part of the design of the simplified business planning and tariff exercise. The study team recommends all the LGU-run WSP partners in the Bohol bulk water supply project to study the demand in their city even as they are currently only supplying a small portion of it together with private water service providers.

The water demand was projected based on the current and projected population, considering all the water users in the area. The tourist population was also considered (Table 20).

Table 20. Population, Household and Tourist Projection for Tagbilaran City

	2024	2030	2035	2040	2045	2050	2055
Projected Population	112,348	123,690	133,082	142,190	151,097	160,562	170,620
Household	24,966	27,487	29,574	31,598	33,577	35,680	37,916
Tourist count	157,074	257,947	390,003	589,669	891,561	1,348,023	1,938,911

Notes:

*Projected population was based on 2020 PSA data.

*Projected households were computed using the following formula: population/average household size of 4.5 for Tagbilaran City

Table 21 below shows the water service connections projections.

Table 21. Water Service Connections Projection for Tagbilaran City

Classification	Water Service Connections (Count)						
	2024*	2030	2025	2040	2025	2050	2025
Residential	24,966	27,487	29,574	31,598	33,577	35,680	37,916
Commercial	1,746	1,786	1,834	1,886	1,942	2,006	2,078
Institutional	119	263	264	264	265	266	267
Total	26,832	29,536	31,671	33,748	35,784	37,952	40,261

Notes:

*2024 values in the table are the projected WSCs if the whole Tagbilaran City is connected to a WSP.

*Actual 2024 water service connections for TCWS total 6,461.

Water demand was projected based on residential, commercial, institutional, and tourist populations. The first step was to estimate the unconstrained demand for the entire city. Unconstrained water demand refers to the volume of water required to serve the target population at the desired service level, assuming no limitations. This estimate accounts for all factors influencing consumption, such as population growth, economic activity, legislation, and policy implementation. It does not, however, consider constraints like pricing, willingness or ability to pay, or even technical limitations (e.g., network capacity or insufficient water sources).

To determine unconstrained demand, the Average Daily Demand (ADD) was calculated. ADD represents the average daily water requirement within the project area. This involved projecting water demand for residential households, commercial establishments, and institutional users, while

also incorporating tourist consumption. The total projected demand, based on ADD, is the sum of domestic, commercial, institutional, and tourist requirements. A summary of these estimates is presented in Table 22 below.

Table 22. Total Demand for Tagbilaran City

Classification	Water Demand (MLD)						
	2024	2030	2035	2040	2045	2050	2055
Residential	20.82	22.92	24.66	26.35	28.00	29.76	31.62
Commercial	5.08	5.20	5.33	5.48	5.65	5.83	6.04
Institutional	1.72	2.37	2.37	2.38	2.39	2.39	2.40
Total	27.62	30.48	32.37	34.21	36.04	37.98	40.07
Tourist	0.28	0.47	0.71	1.06	1.61	2.43	3.50

Assumptions:

- Residential liters per capita per day (lpcd): 185
- Commercial unit consumption: 2.91 cu.m./connection/day
- Institutional unit consumption: 9 cu.m./connection/day
- Tourist lpcd: 197

After estimating the total water demand, NRW is factored in. The volume equivalent to NRW is added to the projected demand to determine the final unconstrained demand for the city. This adjustment ensures that losses from leakage, theft, and other non-billed uses are accounted for.

The assumed NRW target is in Table 23 below:

Table 23. NRW Projection for Tagbilaran City

	2025	2030	2035	2040	2045	2050	2055
% NRW	40%	26%	20%	20%	20%	20%	20%

The resulting unconstrained demand projection for the city is presented in Table 24 below.

Table 24. Total Unconstrained Demand for Tagbilaran City

	2025	2030	2035	2040	2045	2050	2055
Unconstrained demand	39.90	41.77	41.34	44.10	47.06	50.52	54.46

The net demand is calculated as the difference between the projected unconstrained demand and the volume supplied by existing water sources in the city. This figure represents the demand gap that must be met through additional water supply. For this analysis, the following sources were considered as existing supply:

- Water provided by private WSPs namely BWUI and Richli Water

■ Groundwater sources currently used by TCWS

It was also assumed that, once there is an available bulk water supply, TCWS will discontinue deep well operations and transition to bulk water supply from the province. This assumption was incorporated into the net demand calculation (Table 25).

Table 25. Net Demand for Tagbilaran City which will be served by TCWS

	2025	2030	2035	2040	2045	2050	2055
Unconstrained water demand (MLD)	39.86	41.77	41.34	44.10	47.06	50.52	54.46
Private WSPs (MLD)	19.79	23.45	24.99	27.88	29.94	32.42	35.28
Existing waterworks wells (MLD)	8.40	-	-	-	-	-	-
Net water demand (MLD)	11.72	18.33	16.35	16.22	17.12	18.10	19.18

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Pictures



TA study team courtesy call at Tagbilaran City with City Administrator Alvin Luis Acuzar (fourth from right, in white polo)



TA study team meets with BWUI financial and technical staff



Study team meets with Richli Water President Ralph Lim (fourth from the left in black shirt) and General Manager Engr. Alfredo Calinawan (second from the right in black shirt) and the DILG provincial staff



TA study team with Provincial Government of Bohol Governor Aris Aumentado (right side in brown shirt), WB's Senior Water Supply and Sanitation Specialist Lala Fabella (second from right in brown shirt), and DILG provincial and regional staff. Tagbilaran Mayor Jane Yap and WB's Lead Water Supply and Sanitation Specialist Midori Makino joined the meeting virtually.



Tagbilaran City Waterworks revisits its Vision, Mission and Values



*Project team with Tagbilaran City Waterworks composed of technical, financial and customer service staff.
Mayor Jane Yap joined the discussion virtually.*